学号：　　　　　　　　　姓名：

Homework 1

✧	二极管：中文教材 1.3/1.6（共 2 题）
✧	三极管：中文教材 1.8/1.9/1.10/1.12+ 1.X（共 5 题）

1.3. See Figure P1.3, the input voltage is u_i=5 sin(ωt) the diode's working voltage is U_D=0.7 V, ask: please draw the waveform of input and output voltages (u_O~t and u_O~t).

1.3 电路如**图 P1.3** 所示，已知$u_i = 5\sin\omega t\,(V)$，二极管导通电压 U_D=0.7V。试画出u_i与u_o的波形图，并标出幅值。

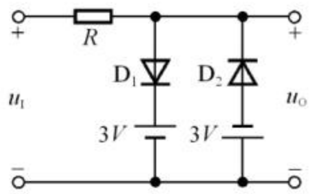

图 P1.3

1.6 As shown in Figure P1.6, voltage regulator (zener) diode's stable voltage is U_Z=6V, the minimum stable current value allowed is I_{Zmin}=5 mA, and the maximum stable current value allowed is I_{Zmax}=25 mA, ask:

(1) If input voltage U_i=10V, then U_0=?

U_i=15V, U_0=?

U_i=35V, U_0=?

(2) If U_i=35 V and load resistor R_L is open, what's going to happen and explain the reason?

1.6 已知**图 Pl.6** 所示电路中稳压管的稳定电压$U_Z = 6V$，最小稳定电流$I_{Z\min} = 5mA$，最大稳定电流$I_{Z\max} = 25mA$。

(1)分别计算U_I为 10V 、15V 、35V 三种情况下输出电压U_O的值；

(2)若$U_I = 35V$时负载开路，则会出现什么现象？ 为什么？

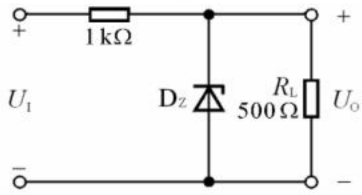

图 P1.6

1.X Please draw the common-emitter and common-base amplifier circuit of PNP transistor when it is in amplification mode. Please also describe its working principle.

1.X 请画出 PNP 晶体管处在放大状态下的共射极与共集极放大电路的接法，并描述工作原理。

1.8 We have two transistors in amplifier circuits as shown in Figure P1.8 (a) and (b). They are in amplification mode. In each transistor, currents from two terminals are known. Ask:

(1) The value and direction of the current from the third terminal;

(2) Draw the symbols of transistors in the circle, and calculate their current amplification factors β.

1.8 现测得放大电路中两只管子两个电极的电流如**图 P1.8** 所示。分别求另一电极的电流，标出其方向，并在圆圈中画出管子，且分别求出它们的电流放大系数β 。

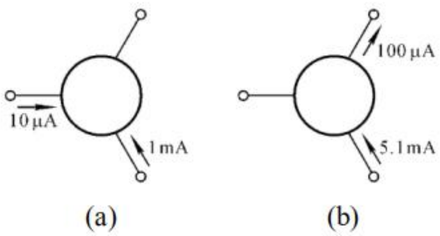

图 P1.8

1.9 Figure P1.9 shows six transistors in six amplifier circuits, transistors are in amplification mode, ask:

(1) Draw the symbols of transistors in the circle;

(2) For each transistor, it is made from silicon or the germanium?

1.9 测得放大电路中六只晶体管的直流电位如**图 P1.9** 所示。在圆圈中画出管子，并说明它们是硅管还是锗管。

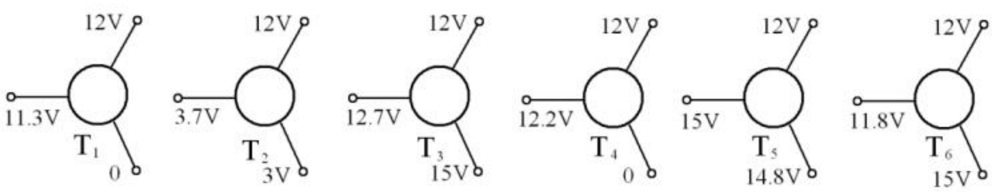

图 P1.9

1.10 See Figure P1.10, we have U_{BE} = 0.7V, β = 50. Ask:
(1) When V_{BB}=0V, please analyze the operation modes/status of the transistor (amplification, cut-off or saturation), and the value of u_o=?
(2) When V_{BB}=1V, please analyze the operation modes/status of the transistor (amplification, cut-off or saturation), and the value of u_o=?
(3) When V_{BB}=3V, please analyze the operation modes/status of the transistor (amplification, cut-off or saturation), and the value of u_o=?

1.10 电路如**图 P1.10** 所示，晶体管导通时 $U_{BE} = 0.7V$，β=50。试分析 V_{BB} 为 0V、1V、3V 三种情况下 T 的工作状态及输出电压 u_O 的值。

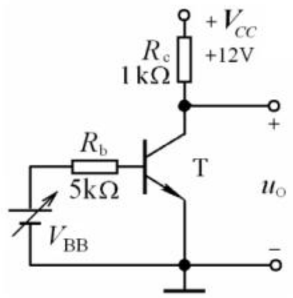

图 P1.10

1.12 For transistors shown in Figure P1.12 (a) to (e), judge whether the transistor is working in the amplification region?

1.12 分别判断**图 P1.12** 所示各电路中晶体管是否有可能工作在放大状态。

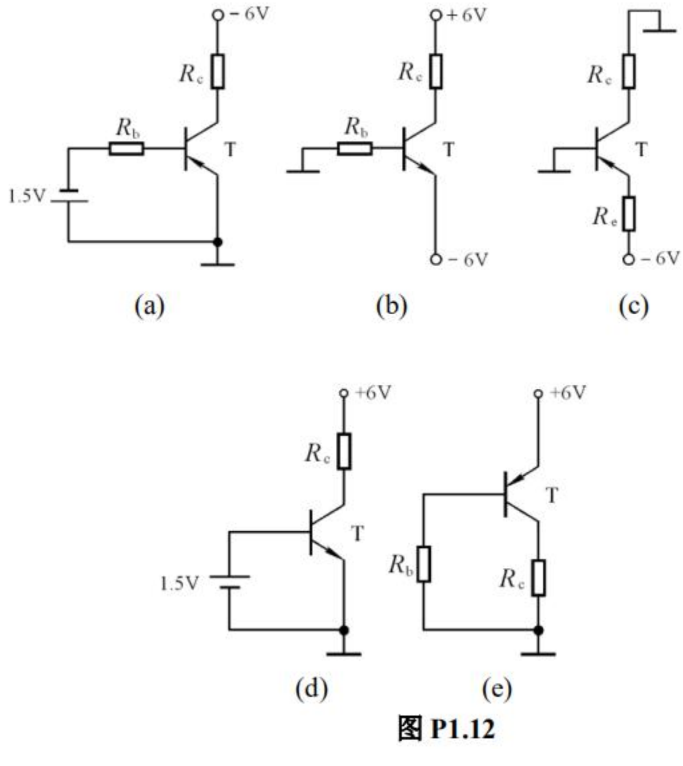

图 P1.12